

Importance of Adopting DfX Principles Early in Design Cycle

With every manufacturer looking for cost reduction while juggling the need to bring newer models to the market, design plays a crucial role. Often, important aspects of design are ignored by the designers, which can lead to substantial cost savings at every stage of the product's lifecycle

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With the explosive growth of electronics in everyday life, one of the major factors that affect all manufacturing companies is their ability to reach the market within a short time-frame. For designers, this means a minimal reworking on their designs. At the same time, with the increasing trend towards outsourced manufacturing, it becomes imperative that products have to be designed in such a way that they ensure the manufacturing becomes easy and cost-optimised.

Most of the time, product designers (especially, when the design is not outsourced) don't worry about the manufacturing aspects while designing the product as the manufacturing is done in-house. By the time the final prototype is ready (which is invariably done without addressing the manufacturing issues), due to market pressure, the product hits the market, based on its existing design. Since the cost-optimising aspects of manufacturing are usually ignored, it leads to cost escalation of the product due to reasons like yield in the manufacturing line, failure in the field, etc.

In most cases, when a new design is moved to the manufacturing line, either the in-house or the contract manufacturer subjects the product to new product introduction (NPI) process, which invariably leads to a design re-spin, resulting in delays as well as downstream additional expenses like

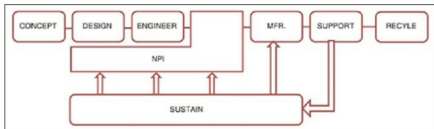


Fig. 1: Lifecycle of a product

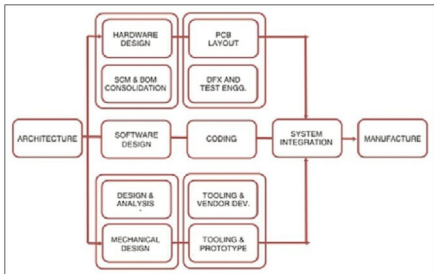


Fig. 2: Conventional development process of a product

recertification charges, etc. One of the key aspects that needs to be looked into by the designers to avoid this wasteful and avoidable expense is 'design for X' or DfX (where 'x' stands for cost, reliability, manufacturing assembly, testing, support, etc.). In this article, we will briefly look at five of the important DfX parameters that impact cost and time-to-market, and how they can effectively reduce the

development time as well as increase the product's profitability.

Anatomy of product's lifecycle

Before we go into the aspects of DfX, let us try to understand the typical product lifecycle, which will enable us to better understand the impact of DfX at each phase of the product's life and how this knowledge can be effectively used